

Multiperiod Planning of Distribution Networks Under Competitive Electricity Market With Penetration of Several Microgrids Part II: Case Study and Numerical Analysis

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Abstract—This paper aims at investigating the effectiveness of the proposed model in Part I. For this, an IEEE 33-bus distribution system is selected, and the proposed algorithm is performed on it. Simulation results obtained from this system show that placing microgrids (MGs) in the distribution network has benefits for all participants; because the proposed model, in addition to considering the objectives of distribution system operator (DSO), finds the potential places to form MGs such that the obtained benefits of MGs are also provided for investments on establishing MGs at these places. The obtained results of this model are applicable for both DSO and MG investors.

Index Terms—Distribution system expansion planning, microgrid (MG) allocation, mixed-integer nonlinear bilevel programming, multimicrogrid planning, multiperiod planning, retail electricity market.

NOMENCLATURE

a, b	Bus index.
P_a/Q_a	Net real/reactive power injection at bus a .
R_{ab}	Resistance of the line between buses a and b .
V, δ	Voltage magnitude, voltage angle.
MG	Microgrid.
DSO	Distribution system operator.
DER	Distributed energy resource.
ESS	Energy storage system.
PV	Photovoltaic panel.
WT	Wind turbine.
FC	Fuel cell.

I. INTRODUCTION

MICROGRIDS (MGs) are new structures in power systems that include distributed energy resources (DERs), storage systems, and loads. Bidirectional behavior of MGs [1], competitive buying/selling electricity market environment, and

existence of different decision makers with somehow contradictory objectives have made their long-term planning a challenging problem. On the other hand, evaluation of the best places for establishing MGs to reduce the losses and improve the voltage profile is another important issue in the planning studies of MGs.

In this two-part paper, multiperiod planning of distribution networks under competitive electricity market with penetration of several MGs is evaluated. In the first part of the paper, a model was presented for multiperiod long-term planning of a distribution network consisting of several MGs with simultaneous consideration of objectives of both the DSO and MG investors under a competitive electricity market environment. The presented model considered the effects of the DSO's decision on the MGs and vice versa and also improved the MG location algorithms. The general scheme of the proposed model is shown in Fig. 1. The problem includes two levels: level one, which is associated with the DSO, deals with the decision making on the location of MGs (to reduce the losses and improve the network voltage) and maximizing the DSO's profit and reliability of the network. The input data for the problem include: grid data such as topology, line capacities, etc., load forecasting for the network, economical parameters such as market price forecasting, cost of load curtailment, interest rate, and to name but a few. The decision making parameters of this level include tariff of selling electricity to the customers and the amount of their load curtailment, the amount of electricity bought from the wholesale market, and the electricity price exchanged with the MGs. Once the price of electricity exchanged by the MGs is determined, then the second level is established. The input data for the second level include: DERs related parameters such as investment and operation costs, nominal power, life time, etc. In this level, the MGs make decisions on several variables to maximize their profit and reliability. The variables include: the investment status of DERS and ESSs, their optimal output power, the amount of load curtailment, and the amount of exchanged power with the DSO. The price and amount of exchanged power between the DSO and MGs are two parameters that model the effects of decision making of the first level on the second level and vice versa. Employing a well-known distribution system, this paper aims at evaluating the effectiveness of the proposed model in Part I. The rest of the paper is organized as follows. The case

Manuscript received May 24, 2017; accepted July 14, 2018. Date of publication July 24, 2018; date of current version April 3, 2019. Paper no. TII-18-1300. (Corresponding author: Asghar Akbari Foroud.)

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Digital Object Identifier 10.1109/TII.2018.2858928

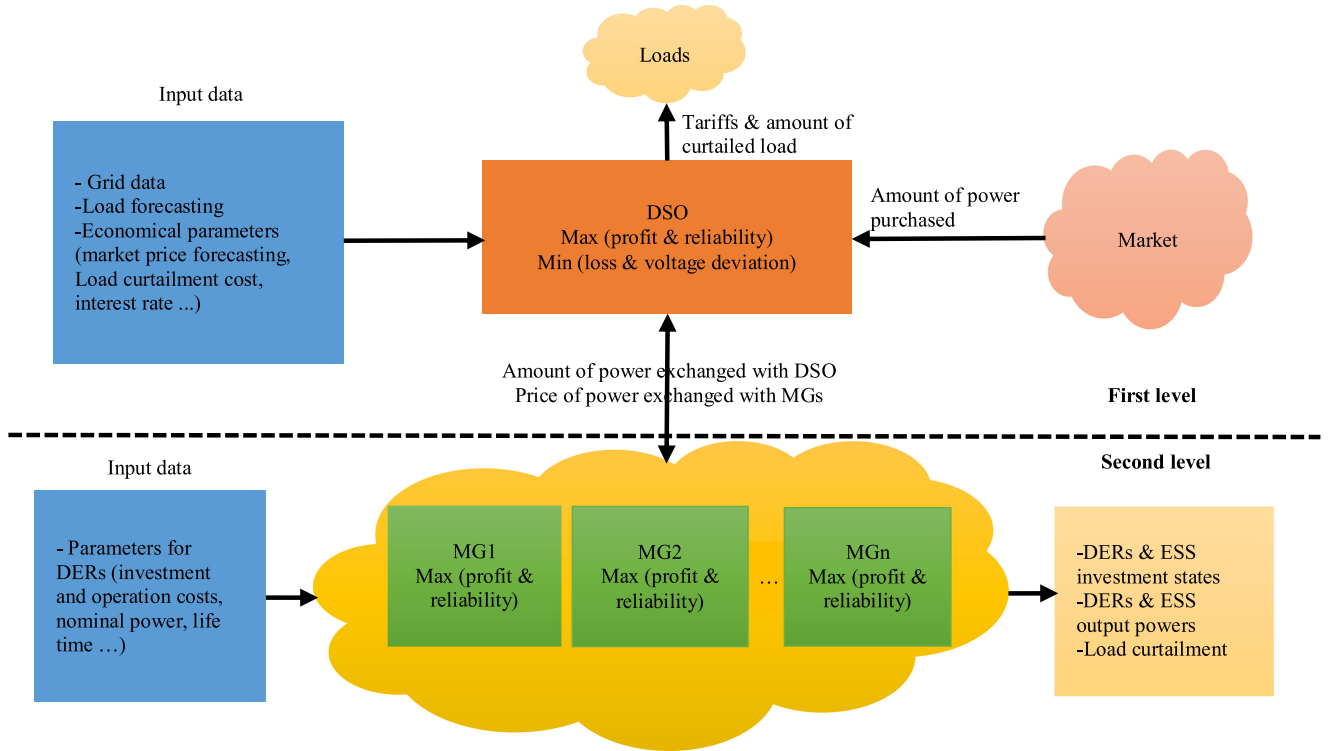


Fig. 1. Planning scheme of multi-MG distribution network under market.

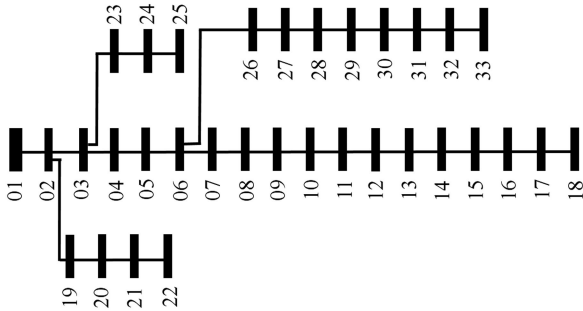


Fig. 2. IEEE 33-bus distribution system.

study on an IEEE 33-bus system is given in Section II. Also, the proposed algorithm is solved for different cases and the results are analyzed in the same section. Conclusions are provided in Section III of the paper.

II. CASE STUDY

In this section, the proposed model in Part I is employed on an IEEE 33-bus radial distribution system depicted in Fig. 2 with minimum load of 3.715 MW and maximum load of 7.957 MW, and its effectiveness is proved. The related data to this distribution system are given in [2] and [3]. The capacity of the line between the distribution network and the upstream network is 9 MW, and the capacity of the energy exchanged between the MGs is assumed to be 2.9 MW. The maximum amount of load curtailment at each MG and distribution network in each year of planning is assumed to be 10% of their total load [4]. Load curtailment costs of the distribution system and MGs at each

year of planning are constant and assumed to be, respectively, 58 and 52 \$/MWh for the first year. The maximum limit for the price of exchanged power between DSO and MGs is 50 \$/MWh [4], and the maximum limit for the tariff of selling electricity to the loads is assumed to be 55 \$/MWh. Annual interest value is set equal to 5% and it is assumed that all monetary parameters during the planning period have annual growth of 6% [5]. Annual growth of load is assumed to be 3% [5]. To consider the investment feedback, the long-term planning horizon is divided into five two-year periods in order to allow the investors adjust their financial policies through benefiting the results of each time period and the algorithm is separately performed for each period [6]. The results of each period are used as input parameters for the next period. The proposed methodology is coded in GAMS software and Solver CPLEX is used to solve the problem. The method is implemented in a computer with Core2Due CPU and 4 GB RAM configuration. MATLAB software was also employed to perform power flow. A series of assumptions are considered in the model as follows.

- 1) The proposed model predicts the extension of resources, but the extension of distribution network lines and feeders are not discussed in the paper.
- 2) After selecting a specific bus to form the MG, that bus along with all its downstream buses are a subset for the established MG."

After power flow, first the desired places of DSO for constructing MG are specified based on the defined index in this paper. System buses are ranked according to loss sensitivity factor (LSF) and voltage sensitivity factor (VSF) and are given, respectively, in Figs. 3 and 4 (peak load in the first year was con-

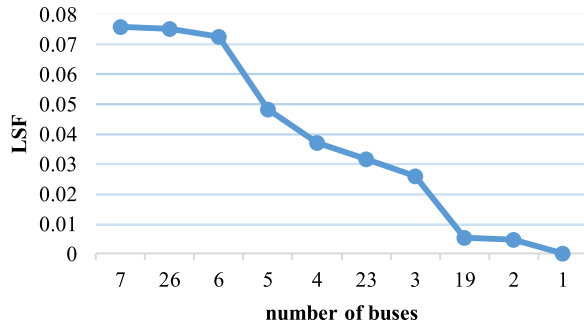


Fig. 3. Bus ranking based on LSF index.

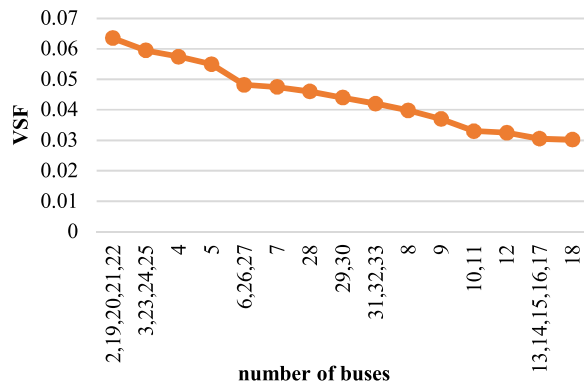
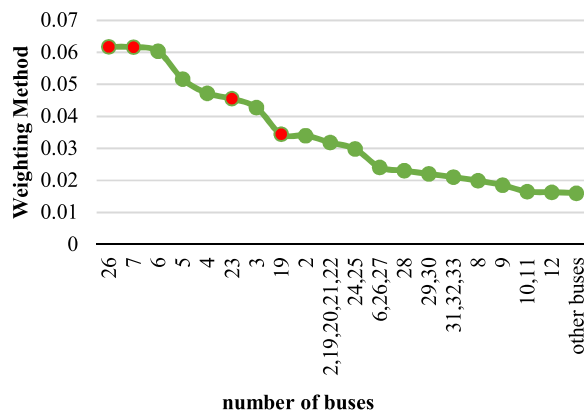


Fig. 4. Bus ranking based on VSF index.

Fig. 5. Bus ranking based on weighting method ($\omega_1 = \omega_2 = 0.5$).

sidered when calculating these factors). Then, based on these figures and using the presented weighting method in Part I, the values are recalculated for each bus and ranked in Fig. 5. Weighting coefficient values can be defined depending on the DSO's opinion. For instance, if loss reduction is of more importance for DSO than improving the voltage profile, then its weighting coefficient should be greater. It is assumed in this paper that both topics are of equal importance for DSO, and therefore the weighting coefficients are equal and it is chosen to be 0.5.

Place selection is performed as such that a bus with the maximum priority along with its subsets form an MG. Therefore, the bus that has already formed an MG is removed from the priority list. Also, the upstream bus of a bus that has already formed an MG, is removed from the list if it has no subsets. Regarding

TABLE I
DATA OF THE CONSIDERED SITES TO FORM AN MG

Site	Buses	Season	Off-peak (KW)	Partial peak (KW)	On peak (KW)
1	7~18	spring	1140	1540	1970
		summer	1310	1800	2300
		autumn	1128	1556	1950
		winter	1075	1480	1860
2	26~33	spring	975	1365	1700
		summer	1122	1550	1960
		autumn	966	1323	1690
		winter	920	1288	1610
3	23~25	spring	985	1360	1720
		summer	1134	1565	1984
		autumn	976	1346	1700
		winter	930	1283	1627
4	19~22	spring	381	533	678
		summer	440	602	770
		autumn	378	521	660
		winter	360	493	630

TABLE II
DERS PARAMETERS

Technology type	Nominal power (KW)	Life time (year)	Maximum installation limit (number)	Investment cost (\$/unit)	Operation cost (\$/KWh)
PV	100	30	15	750000	0.01
FC	350	10	11	1750000	0.03
WT	275	20	15	715000	0.003
DIESEL	375	10	12	257812	0.9
ESS	250	12	10	200000	0.003

these note, there are four points to form an MG from the DSO viewpoint that are buses 26, 7, 23, and 19, respectively. These four buses are considered as points of common couplings. As a result, four sites to form an MG are added to the data of Table I. In this table, low, middle, and peak loads have been presumed to be 8, 12, and 4 h, respectively [6]. The above-mentioned calculations ensure the DSO it has chosen the proper places to reduce the losses and support the distribution system voltage. Selecting the number of sites to establish the MGs depends on the DSO's will, and he makes decisions based on his policy with this regard. In this paper, to show the capabilities and efficacy of the proposed model when faced with multiple MGs, all the four buses with the priorities were selected for establishing MGs.

Every MG requires the installation of DER and ESS resources to supply its present and future loads. The data related to the employed technologies in these MGs are given in Table II [7]–[10]. In this table, the cost related data are given for the starting year of planning.

The buses that do not form MGs include the loads that must be supplied by the distribution system. The amount of these loads in four seasons and three load blocks is provided in Table III.

Every MG makes an effort to plan in such a way that it benefits from maximum autonomy. For this, it considers a reserve in the system as great in size as the biggest employable technology. To clarify the subject, several case studies are taken into account.

Case 1: Distribution system planning without installation of MG.

TABLE III

DISTRIBUTION SYSTEM LOAD DATA IN THE STARTING YEAR OF PLANNING

Season	Off-peak	Partial peak	On peak
Spring	455	637	819
Summer	524	733	943
Autumn	451	631	811
Winter	430	600	774

TABLE IV

WHOLESALE MARKET PRICE OF ELECTRICITY IN THE STARTING YEAR OF PLANNING (\$/KWH)

Season	Off peak	Partial peak	On peak
Spring	0.057	0.063	0.070
Summer	0.074	0.081	0.090
Autumn	0.045	0.050	0.055
Winter	0.041	0.045	0.050

Case 2: Distribution system planning based on MG

Case 3: Making limitations for installation of some technologies in a number of sites (inappropriate installations of some DERs in those site due to technical limitations of the region).

A. Case 1

In this case, it is assumed that there is no investment in the distribution for the installation of DERs or forming MGs. Therefore, in these situations, DSO buys electricity from the wholesale market, and considering its marginal profit sells electricity to the consumers. For better comparison, it is assumed that the desired sites for installing MG have become in form a cluster of loads. Table IV shows the wholesale market price of electricity for different ranges in the first period of the planning. Similar to other economical parameters, it is assumed that the market price during the planning period has 6% annual growth.

The obtained results for Case 1 are summarized in Table V. In this table, cost means the summation of costs at each period. To obtain the rate of changes, the amount of monetary parameters was divided by the amount of annual energy.

According to Table V, in the first period of planning, although summation of maximum load and loss is less than the capacity of line between DSO and upstream network, the load is curtailed. This load curtailment occurs when the wholesale market price is greater than the maximum selling tariff. Therefore, DSO curtails the load and sells energy to the rest of the loads with maximum selling tariff. It is clear from the last period of planning that if there is not a codified planning for establishing the MG or DER resources in the distribution system, then the maximum load of the distribution system (even if the acceptable maximum load curtailment is performed) will be greater than the maximum line capacity between the distribution system and the upstream network (9 MW), so the algorithm cannot converge to an optimal solution. As a result, if DSO has no investments on generation, it must curtail the higher load occurring in this situation. But in case there is no limitation for the load curtailment, as seen from Table V, the load curtailment cost in the last period of planning

experiences a high jump. This cost is due to 19.2% maximum load curtailment

In Table V, the subtraction of summation of DSO incomes (due to selling electricity to consumers) from summation of DSO costs (due to buying electricity from the wholesale market and load curtailment cost) must be exactly equal to DSO profit. However, there is a small difference. The reason for this difference is considering the losses.

B. Case 2

In this case, the candidate buses for Case 1 form an MG, and their planning is performed. It has been assumed that except the maximum limit for the installation of each technology there is no limitation for installing DER resources by MGs. Furthermore, for simplifications, intermittency of PV and WT resources has been neglected. The results of performing algorithm in this case are summarized in Table VI. In this table, the number of installed technologies, cost of buying energy from DSO, cost of selling energy to DSO, and profit of each MG at each period are shown. Profit of MG in this case is defined as: the cost that must be paid to DSO if it was not MG, plus selling to DSO minus MG costs (including installation and operation costs, buying from DSO, and load curtailment). Whenever the price of the wholesale market is lower than DERs generation cost or load curtailment cost, then MGs tend to buy their required electricity from DSO. Whenever the price of wholesale electricity market is higher than load curtailment cost but is lower than DERs generation cost, then MGs curtail a part of the load and supply rest of the load through buying electricity from DSO. When the price of the wholesale market is higher than DERs generation cost but is lower than load curtailment cost, then MGs supply that part of the load that they tend to curtail through DSO, and use DER resources to supply the rest of the loads and sell their surplus electricity to DSO, and this way they obtain profits. When the price of the wholesale market is higher than load curtailment cost and DERs generation costs, then MGs do not tend to buy electricity from DSO and they are most likely to sell electricity. In Table VI, the summation of energy cost and obtained profit at each planning period are listed.

In Table VI, the profit of MG at the first period and some other periods is a negative value. The reason is that at the first period the volume of investments on MGs is high (because of DER technologies establishments), hence the investment cost is high and the overall profit is reduced. Also, each period that the DER technology is installed the profit is impacted. For example, MG1 at period 5 has installed an ESS, so its profit at the end of this period is less than that of the previous period. Or, consider MG2 that has installed a WT at period 5 has resulted in a significant reduction in its profit (due to the high capital investments of DER technology).

Additionally, at two periods the MG has recognized to shed the load instead of installing more technology. Therefore, the cost of load curtailment and buying electricity from the DSO has increased and the income through selling electricity to the DSO has decreased, and this has led to the reduction of profit for that MG (e.g., period 4, MG3, and period 5, MG4).

TABLE V
RESULTS OF PERFORMING THE ALGORITHM IN CASE 1

	Period 1		Period 2		Period 3		Period 4		Period 5	
	Cost of energy consumption (\$)	Peak load (KW)	Cost of energy consumption (\$)	Peak load (KW)	Cost of energy consumption (\$)	Peak load (KW)	Cost of energy consumption (\$)	Peak load (KW)	Cost of energy consumption (\$)	Peak load (KW)
Site 1	561,384.8	2369	595,573.1	2513	631,843.5	2666	670,322.8	2694	711,145.4	3000
Site 2	582,724.5	2018	618,212.4	2141	655,861.5	2271	695,803.5	2291	738,177.9	2556
Site 3	525,938.3	2043	557,967.9	2167	591,948.1	2299	627,997.8	2333	666,242.9	2588
Site 4	203,900.3	793	216,317.8	841	229,491.6	893	243,467.6	927	258,294.8	1005
Other loads	246,541.8	971	261,556.2	1030	277,485.0	1093	294,383.8	1129	312,311.8	1230
Peak load of distribution system (KW)	8194		8692		9222		9374		10379	
Maximum loss (KW)	483.5		521.8		555.7		562.8		609.4	
DSO's profit (\$/MWH)	30.846		32.118		33.354		38.438		41.649	
Cost of load curtailment (\$/MWH)	4.610		5.497		6.552		8.774		12.462	
DSO's payment to wholesale market (\$/MWH)	354.564		376.157		399.065		418.492		439.955	

TABLE VI
MG SITUATIONS AFTER RUNNING THE ALGORITHM IN CASE 2 FOR DIFFERENT PERIODS

MG	Period	ESS	Diesel	WT	FC	PV	Total installed capacity (KW)	Cost of energy purchased from DSO (\$)	Cost of energy sold to DSO (\$)	Profit of MG (\$)
1	1	3	3	2	1	1	2825	103,905.1	242,364.3	-4,904,288.4
	2	3	3	2	1	2	2900	124,269.1	203,133.7	-120,641.3
	3	4	3	2	1	2	3050	148,624.2	307,624.6	542,158.6
	4	4	3	2	1	2	3250	176,862.8	144,322.0	566,584.9
	5	5	3	2	1	2	3425	200,466.7	236,033.5	462,525.8
2	1	2	3	1	1	2	2475	104,763.3	152,458.6	-4,071,002.9
	2	2	3	1	1	3	2575	125,524.5	134,104.2	-177,707.7
	3	3	3	1	1	3	2675	155,407.4	260,739.5	503,517.2
	4	3	3	1	1	3	2825	168,982.0	125,727.0	582,194.9
	5	3	3	2	1	3	2975	189,033.5	261,748.1	12,104.8
3	1	2	3	2	0	3	2475	112,003.0	79,282.7	-4,121,363.8
	2	3	3	2	0	3	2625	103,505.1	56,431.9	495,896.8
	3	3	3	2	0	3	2725	145,270.1	26,285.1	551,210.1
	4	3	3	2	0	3	2725	184,150.2	257,496.6	501,842.9
	5	4	3	2	0	3	2975	206,864.0	113,922.0	315,070.6
4	1	1	1	2	0	0	1250	100,415.1	18,258.1	-2,211,232.8
	2	2	1	2	0	0	1250	132,627.7	241,881.7	122,985.1
	3	2	1	2	0	0	1350	126,602.4	192,074.7	283,807.6
	4	2	1	2	0	0	1350	120,296.6	139,154.7	251,510.0
	5	2	1	2	0	0	1450	147,188.1	82,927.2	177,965.2

Table VII shows the DSO profit and the cost of buying electricity by consumers in the Case 2 for different periods. To obtain the rate of changes, the amount of monetary parameters was divided by the amount of annual energy.

By comparing Tables VI and VII with Table V, some conclusions can be extracted as follows.

- 1) Installation of MG in the distribution system reduces the energy consumption costs for each MG. As it can be observed, the cost of buying energy by an MG from DSO is considerably reduced compared to the case when the same site is considered as a cluster of loads, and this issue encourages the investors to form an MG.

TABLE VII
RESULTS OF RUNNING THE ALGORITHM IN CASE 2

Period	Period 1	Period 2	Period 3	Period 4	Period 5
DSO's profit (\$/MWH)	17.396	18.405	21.557	25.177	23.728
Cost of load curtailment (\$/MWH)	0.265	0.282	0.300	0.318	0.336
DSO's payment to wholesale market (\$/MWH)	64.918	75.158	89.839	100.232	119.182
Cost of energy sold from DSO to other loads (\$/MWH)	5.360	4.719	6.032	6.400	6.789
Total distribution load on peak (KW)	971.3	1030.4	1093.2	1159.8	1230.4
Maximum loss (KW)	311.8	338.6	356.2	380.1	413.9

TABLE VIII
COMPARISON BETWEEN INCOME, COST, AND PROFIT OF DSO IN TWO CASES

Case 1					
Period	Period 1	Period 2	Period 3	Period 4	Period 5
Total DSO's revenue	2,120,489.7	2,249,627.4	2,386,629.7	2,531,975.5	2,686,172.8
Total DSO's expense	1,952,782.9	2,075,004.8	2,205,287.6	2,322,989.3	2,459,731.9
Total DSO's profit	165,772.8	172,535.4	179,119.3	206,695.0	224,003.3
Case 2					
Period	Period 1	Period 2	Period 3	Period 4	Period 5
Total DSO's revenue	450,225.5	511,583.4	608,700.1	685,085.6	780,464.3
Total DSO's expense	354,394.5	410,158.6	490,068.5	546,677.9	649,802.1
Total DSO's profit	94,583.8	100,070.4	117,206.8	136,887.3	129,006.6

- 2) Load curtailment cost has dramatically reduced. Hitherto DSO was responsible for supplying the load of the whole distribution system, but after forming MG and since every MG is designed to work in an autonomous mode, most of the time supplies the load itself and reduces the load curtailment of the distribution system.
- 3) The maximum loss of the system calculated through the following equation is reduced [11]:

$$P_L = \sum_{a=1}^n \sum_{b=1}^n A_{ab} (P_a P_b) \quad (1)$$

where

$$A_{ab} = \frac{R_{ab} \cos(\delta_a - \delta_b)}{v_a v_b}. \quad (2)$$

To calculate the losses at each period, the peak load is considered at the end of each period. The reduced loss represents the efficacy of the applied method in this paper for allocating MG to reduce the losses.

- 4) Power consumption cost of the distribution system consumers compared to a system without MG is reduced. In other words, in the conditions that the price of the electricity bought from MGs is lower than that of the wholesale market, DSO prefers to buy the electricity from MGs and that results in a lower price for selling electricity to the consumers. This shows that the optimal installation

of MG in a distribution system can supply the benefits of the distribution system consumers in addition to the investor.

- 5) Since a considerable part of the distribution system loads, that hitherto DSO was responsible for electrifying them, are now supplied by MGs, therefore the predictive load from DSO perspective in the planning horizon experiences a significant drop which leads to some delays in expanding the lines and feeders of the distribution system. DSO should take this point into account in its policy. Also the following can be inferred from Table VIII.

- 1) The DSO cost in Case 2 is less than that of Case 1. Since this cost is because of DSO's payment to the wholesale market for buying energy, after forming MGs a considerable part of load, that DSO was responsible for supplying it so far, is now supplied by MGs themselves. As a consequence, the amount of energy bought from the wholesale market by DSO is reduced and this in turn reduces DSO's costs.
- 2) DSO's income has reduced compared to Case 1. As it was mentioned before, in Case 1 a part of DSO's profit is due to selling electricity to the prone sites for forming MGs. After formation of MGs, consumers of the distribution system that were directly supplied by DSO itself are reduced and this leads to lower income for DSO compared to the previous case.

TABLE IX
RESULTS OF RUNNING THE ALGORITHM IN CASE 3

MG	Period	ESS	Diesel	WT	FC	PV	Total installed capacity (KW)	Cost of energy purchased from DSO (\$)	Cost of energy sold to DSO (\$)	Profit of MG (\$)
1	1	3	4	0	1	3	2900	131,303.0	180,251.8	-5,036,731.3
	2	3	4	0	1	3	2900	154,937.5	39,877.7	-190,355.2
	3	4	4	0	1	3	3150	184,357.7	136,597.9	286,713.1
	4	4	4	0	1	4	3250	220,328.9	72,014.3	-370,386.0
	5	5	4	0	1	4	3500	257,784.8	155,018.1	240,005.8
2	1	2	4	2	0	0	2550	111,949.4	127,007.2	-4,103,863.1
	2	2	4	2	0	0	2550	133,107.8	57,450.8	-233,555.7
	3	3	4	2	0	0	2800	188,160.6	176,253.5	328,601.7
	4	3	4	2	0	0	2800	222,970.3	41,618.9	373,744.9
	5	4	4	2	0	0	3050	262,436.1	144,467.7	252,634.2
3	1	2	3	2	0	3	2475	109,202.9	153,747.5	-4,007,875.1
	2	3	3	2	0	3	2725	101,331.5	286,138.4	571,052.5
	3	3	3	2	0	3	2725	140,945.7	157,369.3	570,744.0
	4	3	3	2	0	3	2725	179,625.7	118,250.8	564,268.2
	5	3	4	2	0	3	3100	201,278.7	129,579.2	322,430.2
4	1	1	1	2	0	0	1175	95,896.4	53,555.7	-2,160,285.4
	2	2	1	2	0	0	1425	126,128.9	260,671.6	146,641.2
	3	2	1	2	0	0	1425	121,158.5	210,724.8	268,836.5
	4	2	1	2	0	0	1425	114,041.2	157,656.4	283,642.8
	5	2	1	2	0	0	1425	150,469.3	101,271.1	195,624.0

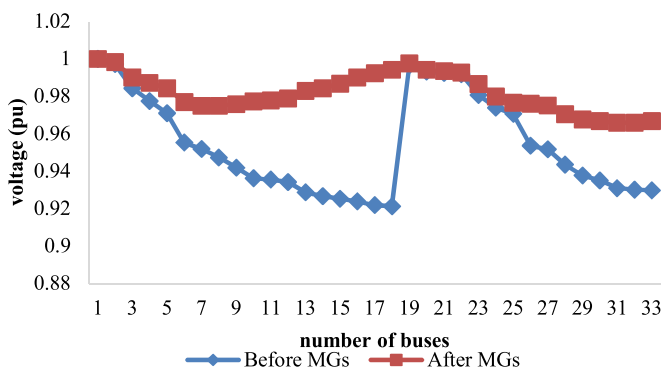


Fig. 6. Distribution system voltage profiles before and after establishment of MGs.

- 3) DSO's profit in the case of installing MGs has reduced compared to that of Case 1. This shows that although by the installation of MGs the incomes and costs of DSO both are reduced, but the rate of income reduction of DSO is greater than its rate of cost reduction. The proposed model is a bilateral game designed based on DSO's profit. In Case 1, all players play a role of consumer. Hence, the number of players that DSO was able to play with to simply obtain income was higher. So, the algorithm had a simple task to approach an optimal solution that provides more profit for the DSO. However, in Case 2, the players behavior changes according to the market data and most of the players can appear as an electricity seller or even do not have any exchanges with DSO. As a result, the resources that DSO merely gains income through them have been degraded and the profit has also been reduced.

Moreover, the rms value of the fundamental components of the network bus voltages before and after the formation of MGs is shown in Fig. 6. The voltages are obtained at each step after running power flow. Backward and forward sweep method was

used for power flow studies in this paper [12]. Harmonic distortions or other distortions produced by distributed generation resources on the voltage waveforms are not dealt with in this paper. The obtained results show that by forming the MGs in the predetermined points of the network, the voltage profile has significantly improved. Hence, the proposed model was somehow successful in clarifying the concerns regarding worsening the voltage profile in the presence of MGs.

C. Case 3

In order to show the capabilities and efficacy of the proposed model, a number of technical limitations for installing DERs in different areas have been taken into account in this case. The technical limitations have impeded the possibility of installing wind and solar resources, respectively, in the sites one and two. As a result, the upper boundary for installing these technologies in the algorithm is assumed to be zero, and the problem is solved again according to this new assumption. The related results are summarized in Table IX.

In addition, the comparison between the cost of bought energy from DSO, the sold energy to DSO, and MG's profits in Cases 2 and 3 is shown in Fig. 7. Only periods 1 and 2 are depicted in this figure. As it is observed, applying limitation by DSO caused the bought energy from DSO in MG 1 and 2 to increase and the sold energy to DSO and its profit to decrease. But in other two MGs this has led to a considerable increase in the profit and the amount of sold energy to the DSO and reduction in the bought energy from DSO. The reason is that the DSO buys more demanded energy by MGs 1 and 2 from MGs 3 and 4. Probably the price of the wholesale market is higher than the electricity price bought from MGs, and it has no economic justification for DSO to buy from the wholesale market.

Table X lists the comparison between different parameters before and after the installation of MGs. One can see that most

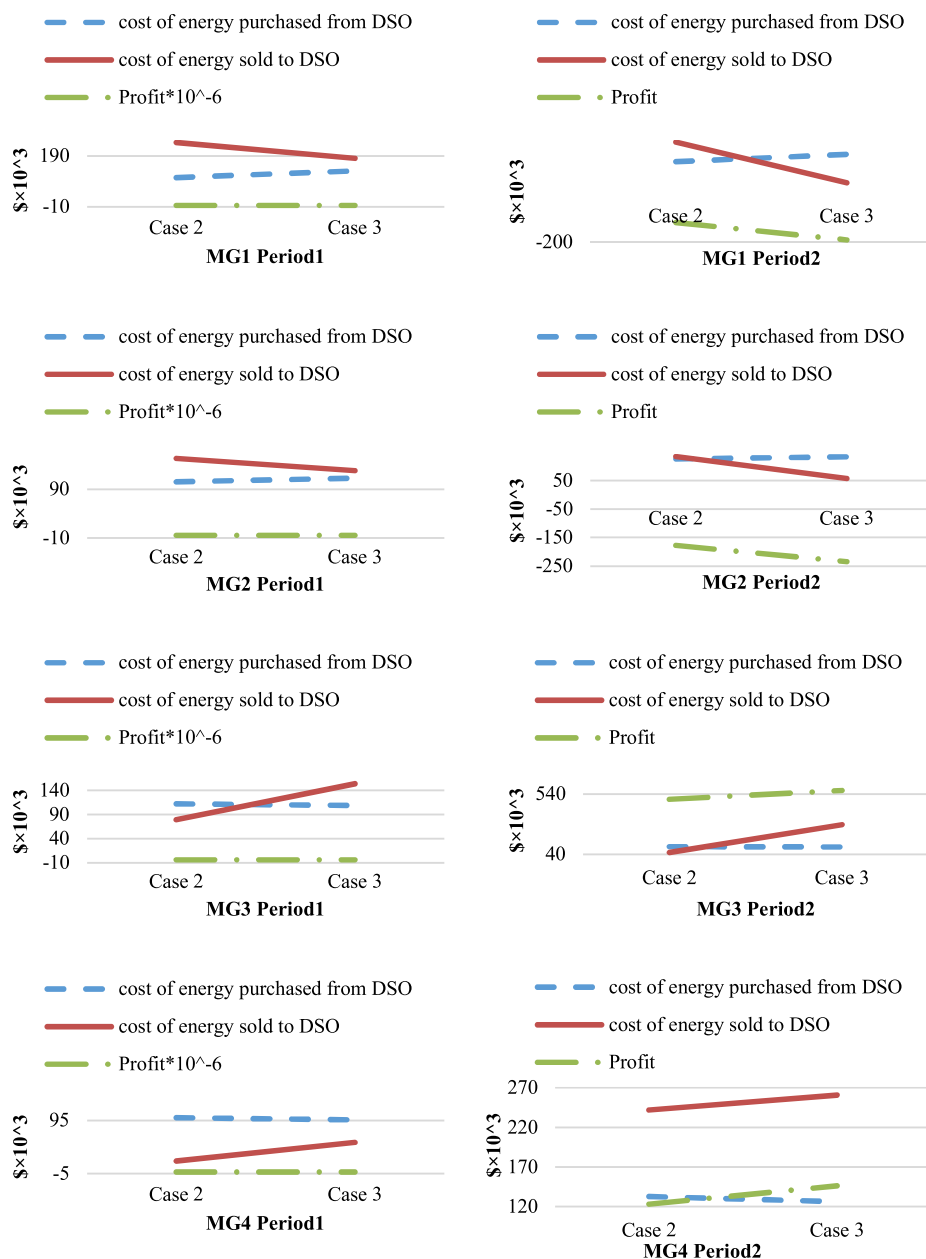


Fig. 7. Comparison between the costs of power bought from DSO and MGs' profits in Cases 2 and 3.

TABLE X
COMPARISON BETWEEN MONETARY PARAMETER BEFORE AND AFTER
MG FORMATION

	Before MG	After MG	Difference%
Cost of Load Curtailment (\$/mwh)	37.9	1.501	96.04
DSO's Payment to Wholesale Market (\$/mwh)	1,988.2	449.33	77.40
Maximum Loss (kw)	2733.2	1800.6	34.12
Average Value of Voltage of Buses (p.u.)	0.954	0.982	2.81
DSO's Profit (\$/mwh)	176.4	106.3	39.76
DSO's revenue (\$/mwh)	2,202.5	558.42	74.65

of the parameters after establishment of MGs have an improving trend. The detailed discussion about this table is presented in the conclusion section.

D. Sensitivity Analysis

Fig. 8 depicts the sensitivity analysis of investment MG1 with respect to the price of a retail market in a planning period. As the price of the retail market is increased, buying electricity from DSO has no economic justification for MG, but selling electricity is justifiable. Consequently, to cut the dependence on DSO and also gain more profit, it starts to invest more on installing technologies. However, the ceiling for its investment is limited based on the available financial resources and its

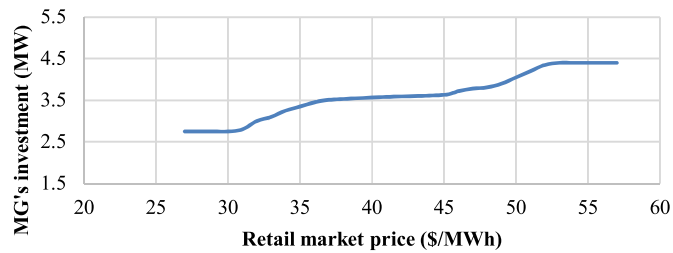


Fig. 8. Change of investments of MG 1 with respect to changes in the price of retail market.

pecuniary policies and hence investment is fixed in a point. On the other hand, as the price of the retail market is reduced, buying energy from DSO is more justifiable for MG than investment. As a result, it starts to reduce investment, but from a point forward the reduced price of the market has no effects on its investment. That is a point where by reducing the investment, MG loses its essential autonomy. As it is clear, decisions of DSO affect the planning of MGs, i.e., MGs react to the DSO's decisions. By defining new contracts to buy energy by MGs in places requiring more load supply, DSO can encourage and entice investors to invest on installing DG technology and system load supply.

III. CONCLUSION

The aim of this set of papers (Part I and II) is to present a model for transition from a traditional distribution network to a network with several MGs under a competitive environment of electricity market. In the proposed model, individual decision makers optimize their objective function without considering those of others, but every decision made by a player interacts mutually with the objective space and decision space of the opposite side. With the aim of earning profit, enhancing reliability, reducing losses, and improving network voltage, DSO is ready to assign some parts of the network (under the title of MG) to the investors. Investors, with the goal of earning profit and enhancing reliability (dictated by DSO), undertake a mutual game with the DSO. It is likely that both sides of the game will not win in all the grounds; however, performing the presented algorithm on an assumed network shows promising results for the investors and operators of the distribution networks. The results (see Table X) maintain that there is an approximate 96% reduction in the load curtailment cost, about 77% reduction in the cost paid by the DSO to the market, about 34% reduction in the losses, and approximately 3% increase in the average bus voltages. Nevertheless, on the contrary, there is an about 40% reduction in the profit for the DSO. As mentioned in the paper, this profit reduction is because of about 75% reduction in the incomes for the DSO. This problem, however, can be justified against some other mentioned advantages. In any case, the DSO is obliged to select between assigning the services to a private section and decentralization, and earning more income and state ownership problems.

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